

CH365 Chemical Engineering Thermodynamics

Lesson 14

Generalized Correlations for Gases and Liquids

Generalized Correlations for Gases Slide 2

Lee-Kesler Method

Byung Ik Lee and Michael Kesler,
AIChE Journal, 1975, 21(3), 511-527

Pitzer Correlation for Z (L13)

(Eq. 3.53)

$$Z = Z^0 + \omega Z^1$$

simple fluids

deviation from
simple fluids:

$$\omega Z^1 = Z - Z^0$$

$$Z^0 = 1 + B^0 \frac{P_r}{T_r}$$

$$Z^1 = B^1 \cdot \frac{P_r}{T_r}$$

(Eq. 3.60)

Lesson 13, Slide 5
formulas for B^0 and B^1
eqns. 3.61 and 3.62

$$Z = 1 + \hat{B} \frac{P_r}{T_r} \text{ (eq. 3.57)}$$

$$\hat{B} = B^0 + \omega B^1 \text{ (eq. 3.59)}$$

$$\begin{aligned} Z &= 1 + (B^0 + \omega B^1) \frac{P_r}{T_r} \\ &= 1 + B^0 \frac{P_r}{T_r} + \omega B^1 \frac{P_r}{T_r} \\ &= Z^0 + \omega Z^1 \text{ (eq. 3.53, implied)} \end{aligned}$$

Lee-Kesler Modification

$Z^{(r)}$: calculated for n-octane

$$Z = Z^0 + \frac{\omega}{\omega^{(r)}} (Z^{(r)} - Z^0) \quad \text{where} \quad Z^1 = \frac{(Z^{(r)} - Z^0)}{\omega^{(r)}}$$

Lee and Kesler used a modified Benedict-Webb-Rubin EOS:

$$Z = 1 + \frac{B}{V_r} + \frac{C}{V_r^2} + \frac{D}{V_r^5} + \frac{c_4}{T_r^3 V_r^2} \left(\beta + \frac{\gamma}{V_r^2} \right) \exp \left(-\frac{\gamma}{V_r^2} \right)$$

B, C, D are functions of T_r
(published in the paper)
 β, γ, c_4 , etc. are constants

Z calculated twice:

Z^0 : calculated for simple fluids

$Z^{(r)}$: calculated for n-octane

Tables: Appendix – Tables D.1-D.4, pp. 676-692

Example: Find Z for n-octane at $P_r=0.4$, $T_r=0.9$

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Table D.1: Values of Z^0 Page 677

$P_r =$	0.0100	0.0500	0.1000	0.2000	0.4000	0.6000	0.8000	1.0000
T_r								
0.30	0.0029	0.0145	0.0290	0.0579	0.1158	0.1737	0.2315	0.2892
0.35	0.0026	0.0130	0.0261	0.0522	0.1043	0.1564	0.2084	0.2604
0.40	0.0024	0.0119	0.0239	0.0477	0.0953	0.1429	0.1904	0.2379
0.45	0.0022	0.0110	0.0221	0.0442	0.0882	0.1322	0.1762	0.2200
0.50	0.0021	0.0103	0.0207	0.0413	0.0825	0.1236	0.1647	0.2056
0.55	0.9804	0.0098	0.0195	0.0390	0.0778	0.1166	0.1553	0.1939
0.60	0.9849	0.0093	0.0186	0.0371	0.0741	0.1109	0.1476	0.1842
0.65	0.9881	0.9377	0.0178	0.0356	0.0710	0.1063	0.1415	0.1765
0.70	0.9904	0.9504	0.8958	0.0344	0.0687	0.1027	0.1366	0.1703
0.75	0.9922	0.9598	0.9165	0.0336	0.0670	0.1001	0.1330	0.1656
0.80	0.9935	0.9669	0.9319	0.8539	0.0661	0.0985	0.1307	0.1626
0.85	0.9946	0.9725	0.9436	0.8810	0.0661	0.0983	0.1301	0.1614
0.90	0.9954	0.9768	0.9528	0.9015	0.7800	0.1006	0.1321	0.1630
0.93	0.9959	0.9790	0.9573	0.9115	0.8059	0.6635	0.1359	0.1664
0.95	0.9961	0.9803	0.9600	0.9174	0.8206	0.6967	0.1410	0.1705
0.97	0.9963	0.9815	0.9625	0.9227	0.8338	0.7240	0.5580	0.1779
0.98	0.9965	0.9821	0.9637	0.9253	0.8398	0.7360	0.5887	0.1844
0.99								
1.00								
1.01								
1.02								
1.05								
1.10								
1.15								
1.20								
1.30								
1.40								
1.50								
1.60								
1.70								
1.80								
1.90								
2.00								
2.20								
2.40								
2.60								
2.80								
3.00								
3.50								
4.00								

$Z^0=0.780$

$Z^1=-0.1118$

$\omega=0.400$

$$Z = Z^0 + \omega Z^1$$
$$= 0.780 + (0.400) \cdot (-0.1118)$$
$$= .7353$$

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Table D.2: Values of Z^1 Page 678

$P_r =$	0.0100	0.0500	0.1000	0.2000	0.4000	0.6000	0.8000	1.0000
T_r								
0.30	-0.0008	-0.0040	-0.0081	-0.0161	-0.0323	-0.0484	-0.0645	-0.0806
0.35	-0.0009	-0.0046	-0.0093	-0.0185	-0.0370	-0.0554	-0.0738	-0.0921
0.40	-0.0010	-0.0048	-0.0095	-0.0190	-0.0380	-0.0570	-0.0758	-0.0946
0.45	-0.0009	-0.0047	-0.0094	-0.0187	-0.0374	-0.0560	-0.0745	-0.0929
0.50	-0.0009	-0.0045	-0.0090	-0.0181	-0.0360	-0.0539	-0.0716	-0.0893
0.55	-0.0314	-0.0043	-0.0086	-0.0172	-0.0343	-0.0513	-0.0682	-0.0849
0.60	-0.0205	-0.0041	-0.0082	-0.0164	-0.0326	-0.0487	-0.0646	-0.0803
0.65	-0.0137	-0.0772	-0.0078	-0.0156	-0.0309	-0.0461	-0.0611	-0.0759
0.70	-0.0093	-0.0507	-0.1161	-0.0148	-0.0294	-0.0438	-0.0579	-0.0718
0.75	-0.0064	-0.0339	-0.0744	-0.0143	-0.0282	-0.0417	-0.0550	-0.0681
0.80	-0.0044	-0.0228	-0.0487	-0.1160	-0.0272	-0.0401	-0.0526	-0.0648
0.85	-0.0029	-0.0152	-0.0319	-0.0715	-0.0268	-0.0391	-0.0509	-0.0622
0.90	-0.0019	-0.0099	-0.0205	-0.0442	-0.1118	-0.0396	-0.0503	-0.0604
0.93	-0.0015	-0.0075	-0.0154	-0.0326	-0.0763	-0.1662	-0.0514	-0.0602
0.95	-0.0012	-0.0062	-0.0126	-0.0262	-0.0589	-0.1110	-0.0540	-0.0607
0.97	-0.0010	-0.0050	-0.0101	-0.0208	-0.0450	-0.0770	-0.1647	-0.0623
0.98	-0.0009	-0.0044	-0.0090	-0.0184	-0.0390	-0.0641	-0.1100	-0.0641
99	-0.0008	-0.0039	-0.0079	-0.0161	-0.0335	-0.0531	-0.0796	-0.0680
10	-0.0007	-0.0034	-0.0069	-0.0140	-0.0285	-0.0435	-0.0588	-0.0879
11	-0.0006	-0.0030	-0.0060	-0.0120	-0.0240	-0.0351	-0.0429	-0.0223
12	-0.0005	-0.0026	-0.0051	-0.0102	-0.0198	-0.0277	-0.0303	-0.0062
15	-0.0003	-0.0015	-0.0029	-0.0054	-0.0092	-0.0097	-0.0032	0.0220
10	0.0000	0.0000	0.0001	0.0007	0.0038	0.0106	0.0236	0.0476
15	0.0002	0.0011	0.0023	0.0052	0.0127	0.0237	0.0396	0.0625
20	0.0004	0.0019	0.0039	0.0084	0.0190	0.0326	0.0499	0.0719
30	0.0006	0.0030	0.0061	0.0125	0.0267	0.0429	0.0612	0.0819
40	0.0007	0.0036	0.0072	0.0147	0.0306	0.0477	0.0661	0.0857
50	0.0008	0.0039	0.0078	0.0158	0.0323	0.0497	0.0677	0.0864
50	0.0008	0.0040	0.0080	0.0162	0.0330	0.0501	0.0677	0.0855
70	0.0008	0.0040	0.0081	0.0163	0.0329	0.0497	0.0667	0.0838
30	0.0008	0.0040	0.0081	0.0162	0.0325	0.0488	0.0652	0.0814
30	0.0008	0.0040	0.0079	0.0159	0.0318	0.0477	0.0635	0.0792
10	0.0008	0.0039	0.0078	0.0155	0.0310	0.0464	0.0617	0.0767
20	0.0007	0.0037	0.0074	0.0147	0.0293	0.0437	0.0579	0.0719
40	0.0007	0.0035	0.0070	0.0139	0.0276	0.0411	0.0544	0.0675
50	0.0007	0.0033	0.0066	0.0131	0.0260	0.0387	0.0512	0.0634
30	0.0006	0.0031	0.0062	0.0124	0.0245	0.0365	0.0483	0.0598
10	0.0006	0.0029	0.0059	0.0117	0.0232	0.0345	0.0456	0.0565
50	0.0005	0.0026	0.0052	0.0103	0.0204	0.0303	0.0401	0.0497
4.00	0.0005	0.0023	0.0046	0.0091	0.0182	0.0270	0.0357	0.0443

Generalized Correlations for Liquids Slide 4

Rackett: $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$ (Eq. 3.68) $Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]}$ (Eq. 3.69)

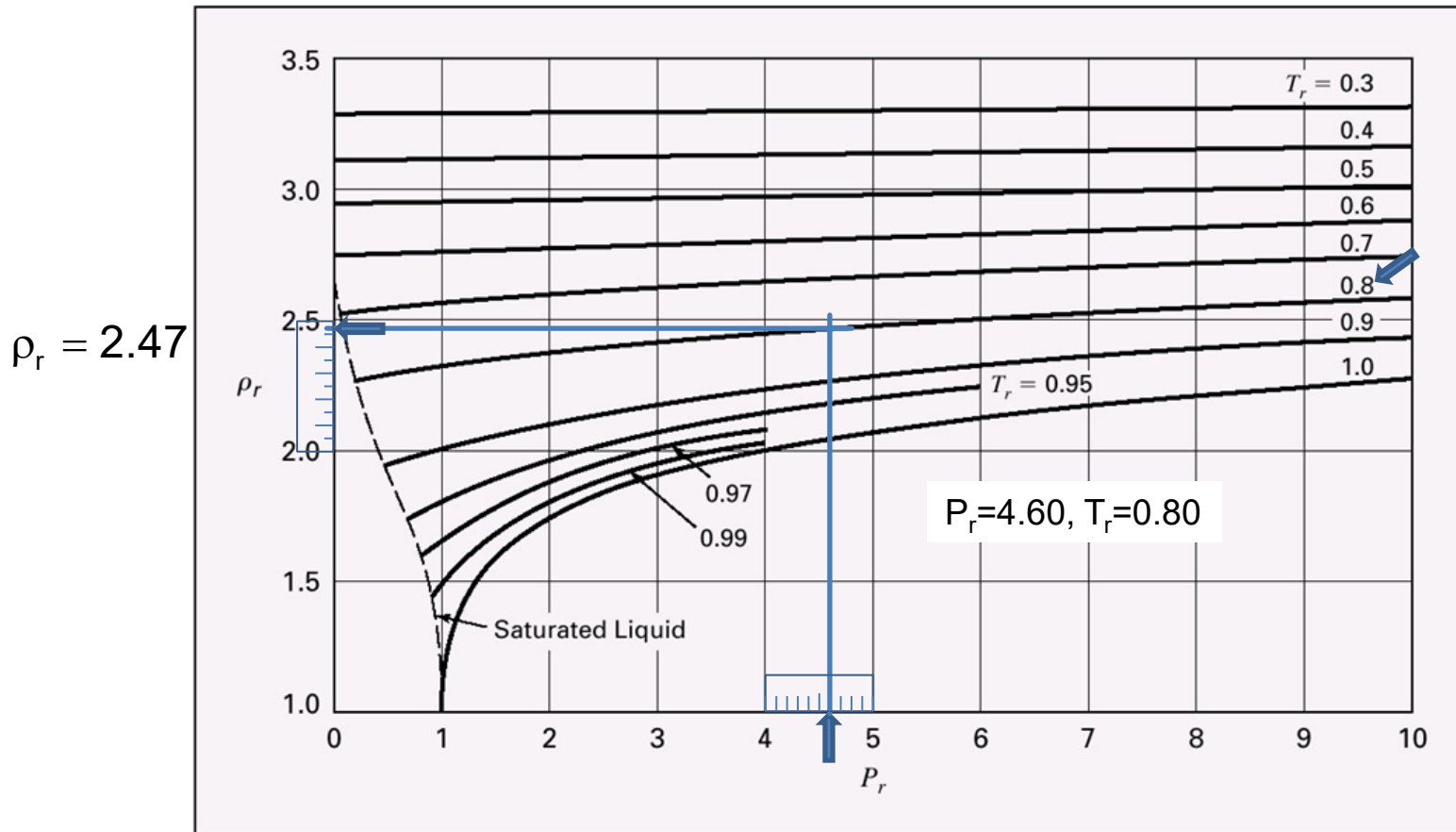


Figure 3.15: Generalized density correlation for liquids.

Lydersen, Greenkorn, and Hougen: $\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V}$ (Eq. 3.70)

Example 3.14

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

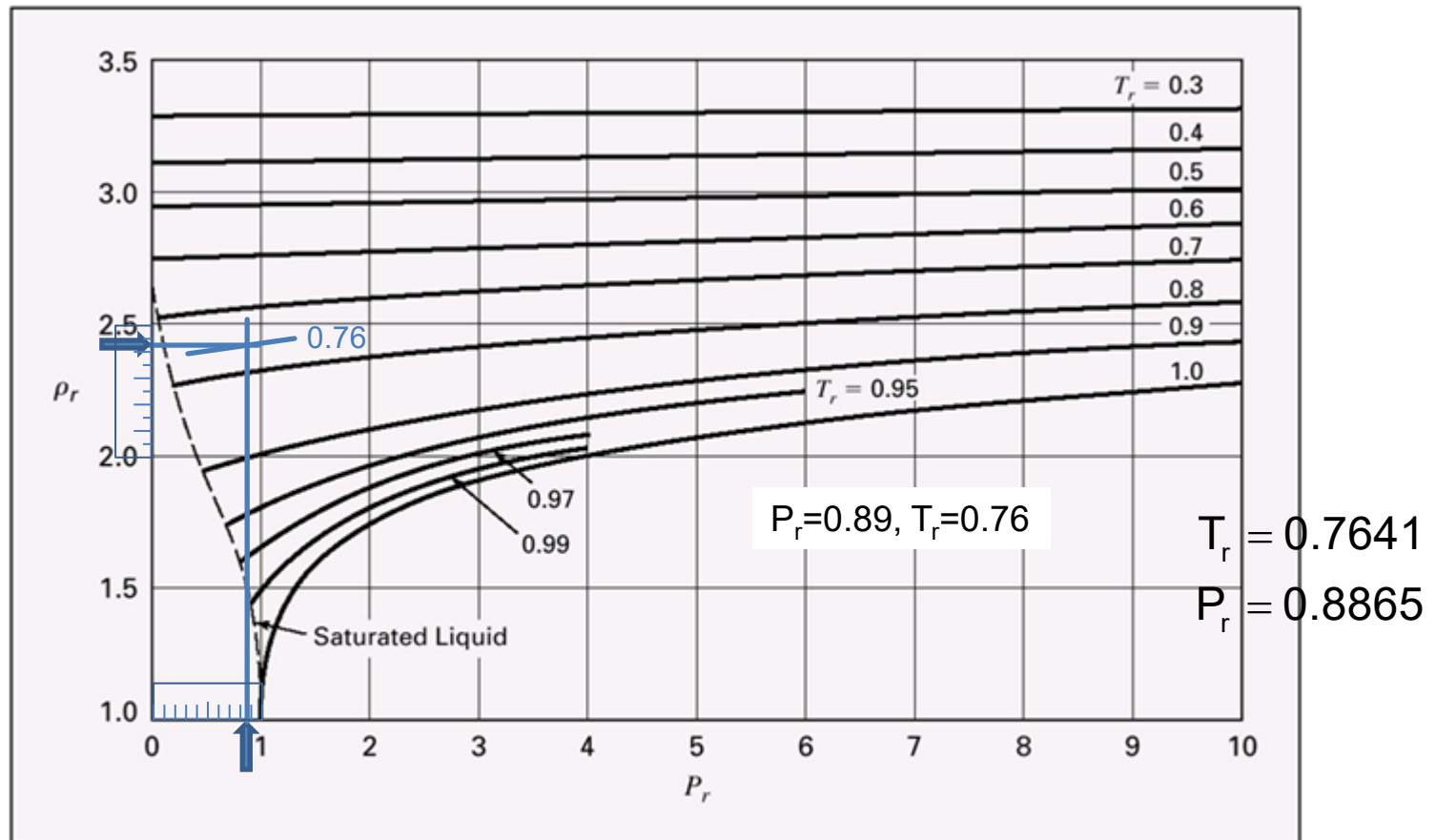
(Liquids)

Example 3.14 , part b, continued

Rackett: $V^{\text{sat}} = V_C Z_C^{(1-T_r)^{2/7}}$ (Eq. 3.68) $Z^{\text{sat}} = \frac{P_r}{T_r} Z_C^{[1+(1-T_r)^{2/7}]}$ (Eq. 3.69)

$$\rho_r = 2.43$$

solution deviates
from book;
authors read
reduced density
as 2.38 on p. 117



Lydersen, Greenkorn, and Hougen:

$$\rho_r \equiv \frac{\rho}{\rho_C} = \frac{V_C}{V} \quad (\text{Eq. 3.70})$$

Example 3.14, continued

For ammonia at 310 K, estimate the molar volume density of (a) the saturated liquid and (b) the liquid at 100 bar.

Questions

Homework

Problem 3.58

To a good approximation, what is the molar volume of ethanol vapor at 480 deg C and 6000 kPa? How does this result compare with the ideal gas?

Answer the problem in four parts:

(a) Lee-Kesler method.

(b) SRK equation.

(c) Ideal gas equation.

For comparison: Compare LK and SRK to IG. If either is less than IG, explain why using knowledge of IG behavior.

Online Bilinear Interpolator Tool for Lee-Kesler Tables:

<https://www.ajdesigner.com/algebra-functions/>